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Tools & Stack: MS SQL Server, Docker, Tableau, Excel, VS Code



PizzaSalesAnalytics&BusinessIntelligence

1.PROJECT OBJECTIVE: To analyze historical transactional data and extract actionable business insights to optimize sales strategy, inventory management, and revenue growth. By leveraging MS SQL Server for data processing and aggregation and Tableau for interactive data visualization, this solution delivers automated key performance indicator (KPI) tracking—including total revenue, average order value, order volume, and unit distribution. Additionally, it models customer purchasing behavior across daily, hourly, and weekly timeframes, evaluates performance across product categories and size preferences, and identifies top- and bottom-performing menu items to support data-driven decision-making.

2.TOOLS USED: Microsoft Excel, Docker (29.6.2 version), MSSQL Server, VS Code, Python (Pandas) Tableau.

3.PROBLEM STATEMENTS:

KPI REQUIREMENTS: We need to analyze key indicators for our pizza sales data to gain insights into our business performance. Specifically, we want to calculate the following metrics:

- 1. Total Revenue:** The sum of the total price of all pizza orders.
- 2. Average order value:** The average amount spent per order, calculated by dividing the total revenue by the total number of orders.
- 3. Total pizzas sold:** The sum of the quantities of all pizzas sold.
- 4. Total Orders:** The total number of orders placed.
- 5. Average pizzas per order:** The average number of pizzas sold per order, calculated by dividing the total number of pizzas sold by the total number of orders.

CHARTS REQUIREMENTS: we would like to visualize various aspects of our pizza sales data to gain insights and understand key trends. We have identified the following requirements for creating charts.

- 1. Daily Trend for Total Orders:** create a bar chart that displays the daily trend of total orders over a specific time period. This chart will help us identify any patterns or fluctuations in order volumes on a daily basis.

2. Hourly Trend for Total Pizzas Sold: create a stacked bar chart that displays the hourly trend of total orders over a specific time period. This chart will help us identify any patterns or fluctuations in order volumes on an hourly basis.

3. Weekly Trend for Total Orders: create a line chart that illustrates the weekly trend of total orders throughout the year. This chart will allow us to identify peak weeks or periods of high order activity.

4. Percentage of sales by pizza category: create a pie chart that shows the distribution of sales across different pizza categories. This chart will provide insights into the popularity of various pizza categories and their contribution to overall sales.

5. Percentage of sales by pizza size: Generate a pie chart that represents the percentage of sales attributed to different pizza sizes. This chart will help us understand customer preferences for pizza sizes and their impact on sales.

6. Total pizzas sold by pizza category: create a butterfly chart that presents the total number of pizzas sold for each pizza category. This chart will allow us to compare the sales performance of different pizza categories.

7. Top 5 Best sellers by Revenue, Total quantity, and Total orders: create a bar chart highlighting the Top 5 best –selling pizzas based on the revenue, total quantity, total orders. This chart will help us identify the most popular pizza options.

8. Bottom 5 Best sellers by Revenue, Total quantity, and Total orders: create a bar chart showcasing the bottom 5 worst selling pizzas based on the revenue, total quantity, total orders. This chart will enable us to identify underperforming or less popular pizza options.

4.DOCKER SET UP:

➤ **Start and Connect the Container:** Launch the SQL Server container in Docker Desktop, then connect to the database instance using VS Code.

➤ **Copy the CSV into Docker (Terminal):**

```
docker cp
```

```
"/path/to/your/pizza_sales.csv<container_name> :/var/opt/mssql/data/pizza_sales.csv
```

➤ **Verify the file inside the container (Terminal):**

```
docker exec -it <container_name> ls -l
```

```
/var/opt/mssql/data/pizza_sales.csv
```

5.DATABASE SETUP:

5.1 DATABASE CREATION:

```
CREATE DATABASE PizzaDB;
```

```
GO
USE PizzaDB;
GO
```

5.2 TABLE CREATION:

```
USE PizzaDB;
GO
DROP TABLE IF EXISTS pizza_sales;
GO
CREATE TABLE pizza_sales
(
    pizza_id INT,
    order_id INT,
    pizza_name_id VARCHAR (100),
    quantity INT,
    order_date_raw VARCHAR (50),
    order_time_raw VARCHAR (50),
    unit_price DECIMAL (10,2),
    total_price DECIMAL (10,2),
    pizza_size VARCHAR (50),
    pizza_category VARCHAR (50),
    pizza_ingredients VARCHAR(MAX),
    pizza_name VARCHAR(MAX)
);
GO
```

5.3 DATA INGESTION & BULK INSERT:

```
USE PizzaDB;
GO
BULK INSERT pizza_sales
FROM 'var/opt/mssql/data/pizza_sales.csv'
WITH
(
    FIRSTROW = 2,
    FIELDTERMINATOR = '|',
    ROWTERMINATOR = '0x0a',
    TABLOCK
);
GO
```

5.4 DATA CLEANING:

```
ALTER TABLE pizza_sales
ADD order_date DATE,
order_time TIME;
GO
-- Convert raw date and time values
```

```

UPDATE pizza_sales
SET
order_date = TRY_PARSE (order_date_raw AS DATE USING 'en-GB'),
order_time = TRY_CAST (order_time_raw AS TIME);
GO
-- Remove temporary raw columns
ALTER TABLE pizza_sales
DROP COLUMN order_date_raw, order_time_raw;
GO

```

6. SQL QUERIES:

6.1 BASIC & EXPLORATORY QUERIES:

```

SELECT * FROM pizza_sales;
SELECT count (*) FROM pizza_sales;

```

Output:

Total_Sales
1 48620

```

SELECT TOP 10 * FROM pizza_sales;

```

6.2 KEY PERFORMANCE INDICATORS (KPIs) ANALYSIS:

1. TOTAL REVENUE:

```

SELECT SUM (total_price) AS Total_Revenue FROM pizza_sales;

```

Output:

Total_Revenue
1 817860.05

2. AVERAGE ORDER VALUE:

```

SELECT SUM (total_price)/COUNT (DISTINCT order_id) AS Average_Order_Value
FROM pizza_sales;

```

Output:

Average_Order_Value
1 38.307262

3. TOTAL PIZZAS SOLD:

```

SELECT SUM (quantity) AS Total_Pizzas_Sold FROM pizza_sales;

```

Output:

Total_Pizzas_Sold
1 49574

4. TOTAL ORDERS:

```

SELECT COUNT (DISTINCT order_id) AS Total_Orders FROM pizza_sales;

```

Output:

Total_Orders
1 21350

5. AVERAGE PIZZAS PER ORDER:

```
SELECT CAST (CAST (SUM (quantity) AS DECIMAL (10,2)) / CAST (COUNT
(DISTINCT order_id) AS DECIMAL (10,2)) AS DECIMAL (10,2)) AS
Average_Pizzas_Per_Order
FROM pizza_sales;
```

Output:

Average_Pizzas_Per_Order
2.32

6.3 CATEGORY PERFORMANCE & SALES DISTRIBUTION:

1. DAILY TREND FOR TOTAL ORDERS:

```
SELECT DATENAME (DW, order_date) AS Order_day, COUNT (DISTINCT order_id)
AS Total_Orders
FROM pizza_sales
GROUP BY DATENAME (DW, order_date);
```

Output:

Order_day	Total_Orders
1 Saturday	3158
2 Wednesday	3024
3 Monday	2794
4 Sunday	2624
5 Friday	3538
6 Thursday	3239
7 Tuesday	2973

2. HOURLY TREND FOR TOTAL PIZZAS SOLD:

```
SELECT DATEPART (HOUR, order_time) AS Order_Hour, SUM (quantity) AS
Total_Pizzas_Sold
FROM pizza_sales
GROUP BY DATEPART (HOUR, order_time)
ORDER BY DATEPART (HOUR, order_time);
```

Output:

Order_Hour	Total_Pizzas_Sold
1 9	4
2 10	18
3 11	2728
4 12	6776
5 13	6413
6 14	3613
7 15	3216
8 16	4239
9 17	5211
10 18	5417
11 19	4486
12 20	3534
13 21	2545
14 22	1386
15 23	68

3.WEEKLY TREND FOR TOTAL PIZZAS SOLD:

```
SELECT DATEPART (ISO_WEEK, order_date)AS Week_number, YEAR(order_date) AS
Order_Year, COUNT (DISTINCT order_id) AS Total_Orders FROM pizza_sales
GROUP BY DATEPART (ISO_WEEK, order_date), YEAR (order_date)
ORDER BY DATEPART (ISO_WEEK, order_date), YEAR (order_date);
```

Output:

Week_number	Order_Year	Total_Orders			
1 1	2015	254	32 32	2015	426
2 2	2015	427	33 33	2015	435
3 3	2015	480	34 34	2015	407
4 4	2015	415	35 35	2015	394
5 5	2015	436	36 36	2015	397
6 6	2015	422	37 37	2015	435
7 7	2015	423	38 38	2015	423
8 8	2015	393	39 39	2015	288
9 9	2015	489	40 40	2015	433
10 10	2015	420	41 41	2015	334
11 11	2015	494	42 42	2015	386
12 12	2015	416	43 43	2015	352
13 13	2015	427	44 44	2015	371
14 14	2015	433	45 45	2015	394
15 15	2015	488	46 46	2015	400
16 16	2015	414	47 47	2015	392
17 17	2015	437	48 48	2015	491
18 18	2015	423	49 49	2015	424
19 19	2015	399	50 50	2015	417
20 20	2015	458	51 51	2015	430
21 21	2015	414	52 52	2015	298
22 22	2015	398	53 53	2015	171
23 23	2015	423			
24 24	2015	418			
25 25	2015	419			
26 26	2015	416			
27 27	2015	474			
28 28	2015	417			
29 29	2015	428			
30 30	2015	433			
31 31	2015	419			

4. PERCENTAGE OF TOTAL SALES BY PIZZA CATEGORY:

```
SELECT pizza_category, cast(sum(total_price) AS decimal (10,2)) AS total_revenue,
cast(sum(total_price) * 100 / (SELECT SUM (total_price) FROM pizza_sales) AS decimal
(10, 2)) AS PCT FROM Pizza_sales
GROUP BY pizza_category;
```

Output:

	pizza_category	total_revenue	PCT
1	Classic	220053.10	26.91
2	Chicken	195919.50	23.96
3	Veggie	193690.45	23.68
4	Supreme	208197.00	25.46

NOTE:

If you want to apply the month, Quarter, Week filters to the above queries you can use WHERE Clause. Follow some of below examples:

```
SELECT DATENAME (DW, order_date) AS order_day, COUNT (DISTINCT order_id)
AS Total_Orders
FROM pizza_sales
WHERE MONTH (order_date) = 1
GROUP BY DATENAME (DW, order_date);
```

*Here MONTH (order_date) = 1 indicates that the output is for the month of january.

MONTH (order_date) = 4. indicates output for the month of April.

```
SELECT DATENAME (DW, order_date) AS order_day, COUNT (DISTINCT order_id)
AS Total_Orders
FROM pizza_sales
WHERE DATEPART (Quarter, order_date) = 1
GROUP BY DATENAME (DW, order_date);
```

*Here DATEPART (Quarter, order_date) = 1 indicates that the output is for the Quarter 1

MONTH (order_date) = 3 indicates output for Quarter 3.

5. PERCENTAGE OF TOTAL SALES BY PIZZA SIZE:

```
SELECT pizza_size,Cast(sum(total_price) AS DECIMAL (10,2)) AS total_revenue,Cast
(sum(total_price)*100/ (SELECT SUM (total_price) FROM pizza_sales) AS DECIMAL (10,
2)) AS PCT FROM Pizza_sales
```

GROUP BY pizza_size
ORDER BY PCT DESC;

Output:

	pizza_size ↕ ▾	total_revenue ↕ ▾	PCT ↕ ▾
1	L	375318.70	45.89
2	M	249382.25	30.49
3	S	178076.50	21.77
4	XL	14076.00	1.72
5	XXL	1006.60	0.12

6.4 TOP & BOTTOM PERFORMING PRODUCTS:

1.TOP 5 BEST SELLERS BY REVENUE, TOTAL QUANTITY AND TOTAL ORDERS:

TOP 5 BEST SELLERS BY REVENUE:

SELECT TOP 5 pizza_name, SUM (total_price) AS Total_Revenue FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC;

Output:

	pizza_name ↕ ▾	Total_Revenue ↕ ▾
1	The Thai Chicken Pizza↵	43434.25
2	The Barbecue Chicken Pizza↵	42768.00
3	The California Chicken Pizza↵	41409.50
4	The Classic Deluxe Pizza↵	38180.50
5	The Spicy Italian Pizza↵	34831.25

BOTTOM 5 BEST SELLERS BY REVENUE:

SELECT TOP 5 pizza_name, SUM (total_price) AS Total_Revenue FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue ASC;

Output:

	pizza_name ↕ ▾	Total_Revenue ↕ ▾
1	The Brie Carre Pizza↵	11588.50
2	The Green Garden Pizza↵	13955.75
3	The Spinach Supreme Pizza↵	15277.75
4	The Mediterranean Pizza↵	15360.50
5	The Spinach Pesto Pizza↵	15596.00

TOP 5 BEST SELLERS BY QUANTITY:

SELECT TOP 5 pizza_name, SUM (quantity) AS Total_Quantity FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity DESC;

Output:

	pizza_name ↕ ▾	Total_Quantity ↕ ▾
1	The Classic Deluxe Pizza↵	2453
2	The Barbecue Chicken Pizza↵	2432
3	The Hawaiian Pizza↵	2422
4	The Pepperoni Pizza↵	2418
5	The Thai Chicken Pizza↵	2371

BOTTOM 5 BEST SELLERS BY QUANTITY:

SELECT TOP 5 pizza_name, SUM (quantity) AS Total_Quantity FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity ASC;

Output:

	pizza_name ↕ 🔍	Total_Quantity ↕ 🔍
1	The Brie Carre Pizza↵	490
2	The Mediterranean Pizza↵	934
3	The Calabrese Pizza↵	937
4	The Spinach Supreme Pizza↵	950
5	The Soppressata Pizza↵	961

TOP 5 BEST SELLERS BY ORDERS:

```
SELECT TOP 5 pizza_name, COUNT (DISTINCT order_id) AS Total_Orders FROM
pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders DESC;
```

Output:

	pizza_name ↕ 🔍	Total_Orders ↕ 🔍
1	The Classic Deluxe Pizza↵	2329
2	The Hawaiian Pizza↵	2280
3	The Pepperoni Pizza↵	2278
4	The Barbecue Chicken Pizza↵	2273
5	The Thai Chicken Pizza↵	2225

BOTTOM 5 BEST SELLERS BY ORDERS:

```
SELECT TOP 5 pizza_name, COUNT (DISTINCT order_id) AS Total_Orders FROM
pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders ASC;
```

Output:

	pizza_name ↕ 🔍	Total_Orders ↕ 🔍
1	The Brie Carre Pizza↵	480
2	The Mediterranean Pizza↵	912
3	The Calabrese Pizza↵	918
4	The Spinach Supreme Pizza↵	918
5	The Chicken Pesto Pizza↵	938